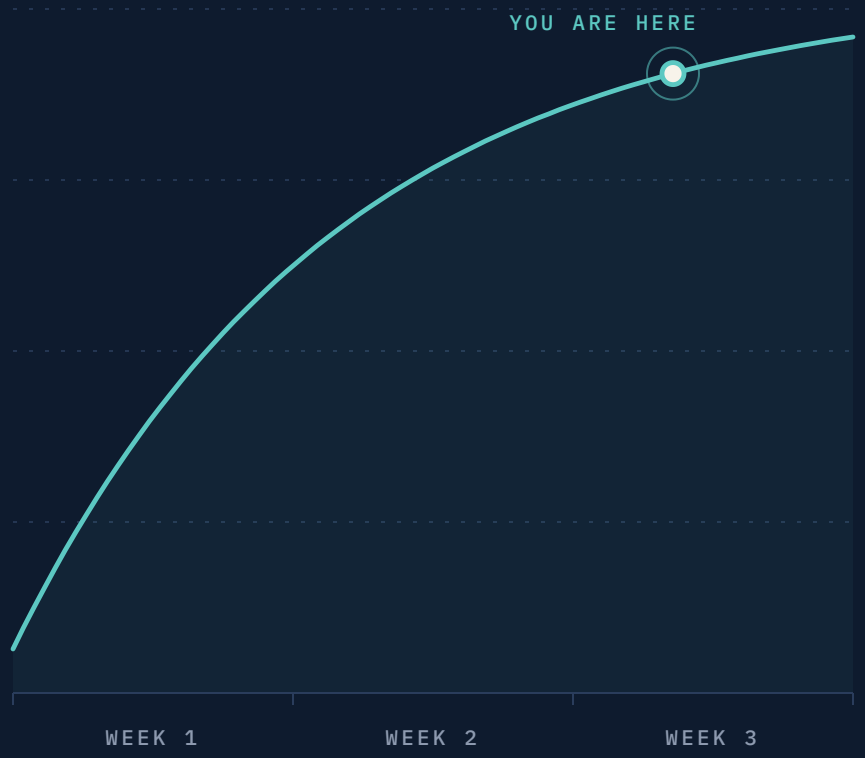
WEEK 3 · DAY 2 · OVERVIEW

Scaling and K-Means



Created by **Jeff Lenamon**

ABOUT 50 MIN WITH EXERCISES / 11 SECTIONS / 4 EXERCISES

This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

Problem Statement

BUSINESS CONTEXT

A credit team wants a quick map of the 821-bond benchmark: which bonds look alike on spread, duration, coupon and rating.

No column in the file says what the groups should be, so there is no target today.

The model has to find the groups on its own: this is **unsupervised learning**.

The bonds and issuers are invented for this course.

OBJECTIVES

Group the bonds with K-means, show that the columns' units can decide the groups and fix that by scaling, choose the number of clusters with the elbow and the silhouette, and describe each cluster in the file's own units.

ABOUT 50 MIN WITH EXERCISES / 11 SECTIONS / 4 EXERCISES

Your Route Through Day 2

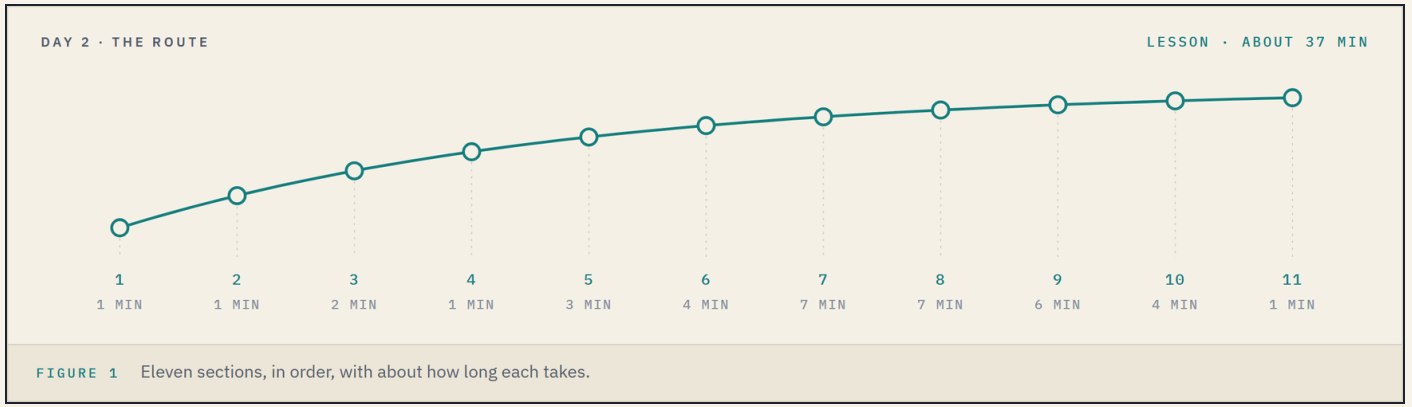


FIGURE 1 Eleven sections, in order, with about how long each takes.

	SECTION	ABOUT
1	Setting Up	1 min
2	Loading the Data	1 min
3	Data Overview	2 min
4	Exploratory Data Analysis	1 min
5	Distance Between Two Bonds	3 min
6	K-Means on the File's Units	4 min

	SECTION	ABOUT
7	Scaling the Columns	7 min
8	Inside the Loop	7 min
9	Choosing the Number of Clusters	6 min
10	Describing the Clusters	4 min
11	Insights and Conclusion	1 min

Time. About 37 minutes for the lesson, then about 12 for the four exercises.

Fix the seed and load the libraries.

Setting Up

FIX THE SEED AND FIND THE DATA

```
import os
os.environ["OMP_NUM_THREADS"] = "1"
os.environ["OPENBLAS_NUM_THREADS"] = "1"
SEED = 42
DATA = "../../../data/"
```

LOAD THE LIBRARIES

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px
from scipy.spatial.distance import pdist
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
```

Run this cell and move on: it fixes the seed and finds the course data.

Loading the Data

WHAT THE FILE HOLDS

- **cusip**: the bond's identifier
- **issuer**: the issuer's name
- **rating**: credit rating, AAA to CCC
- **oas**: option-adjusted spread, bp
- **duration**: modified duration, years
- **coupon**: annual coupon, % of par
- **amount_outstanding**: face value outstanding, dollars (Exercise 1)

READ THE BONDS AND ADD LOG OAS

```
bonds = pd.read_csv(DATA + "benchmark.csv")
ratings = pd.read_csv(DATA + "ratings.csv")
data = bonds.merge(ratings[["rating", "score", "grade"]], on="rating")
data["log_oas"] = np.log(data["oas"])
```

Check the rows, the spreads, and missing values.

Data Overview

THE FIRST FIVE BONDS

	ISSUER	RATING	OAS	COUPON
0	Quorvane Energy	BBB	83	4.875
1	Quorvane Energy	BBB	122	5.750
2	Quorvane Energy	BBB	123	5.000
3	Quorvane Energy	BBB	88	3.875
4	Quorvane Energy	BBB	127	5.375

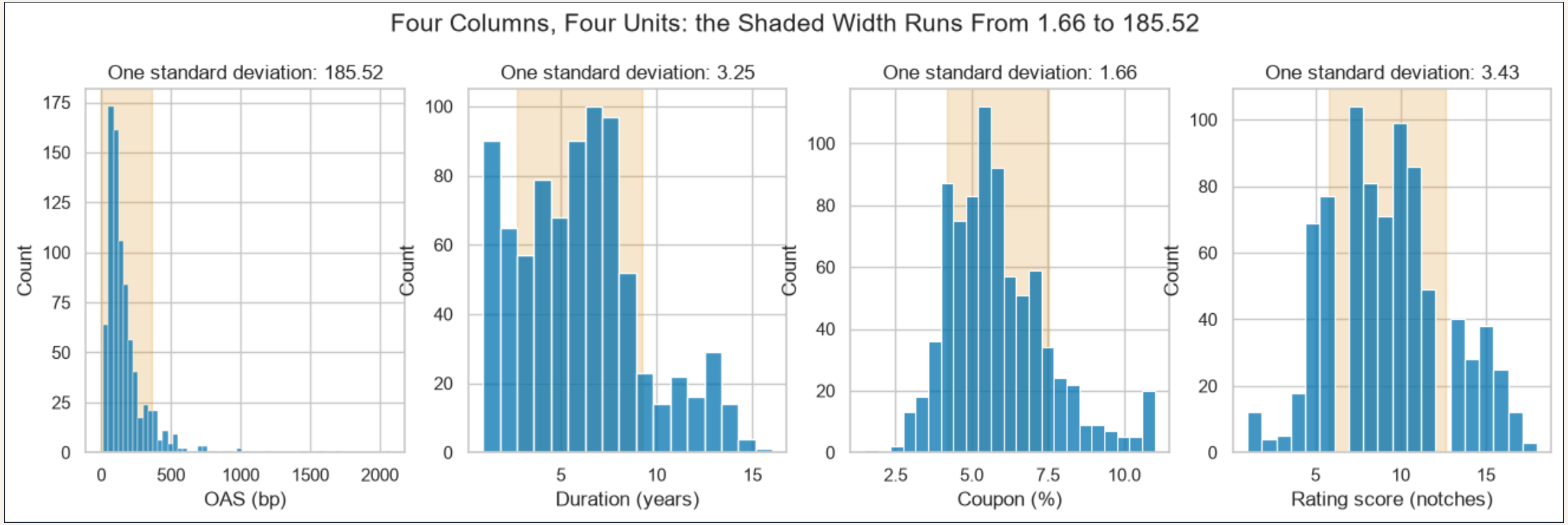
COUNT, MEAN, SPREAD AND RANGE

	OAS	LOG_OAS	DURATION	COUPON	SCORE
count	821.00	821.00	821.00	821.00	821.00
mean	179.08	4.89	5.98	5.85	9.25
std	185.52	0.74	3.25	1.66	3.43
min	15.00	2.71	0.92	1.50	1.00
25%	79.00	4.37	3.43	4.75	7.00
50%	125.00	4.83	5.92	5.62	9.00
75%	210.00	5.35	7.74	6.75	11.00
max	2074.00	7.64	16.04	11.00	18.00

The same clean 821 bonds as week 1, but their columns live on very different scales.

See each column in its own unit.

Exploratory Data Analysis



There is no target column today. The question is which bonds sit close together, so we need a distance.

Measure how far apart two bonds sit.

Distance Between Two Bonds

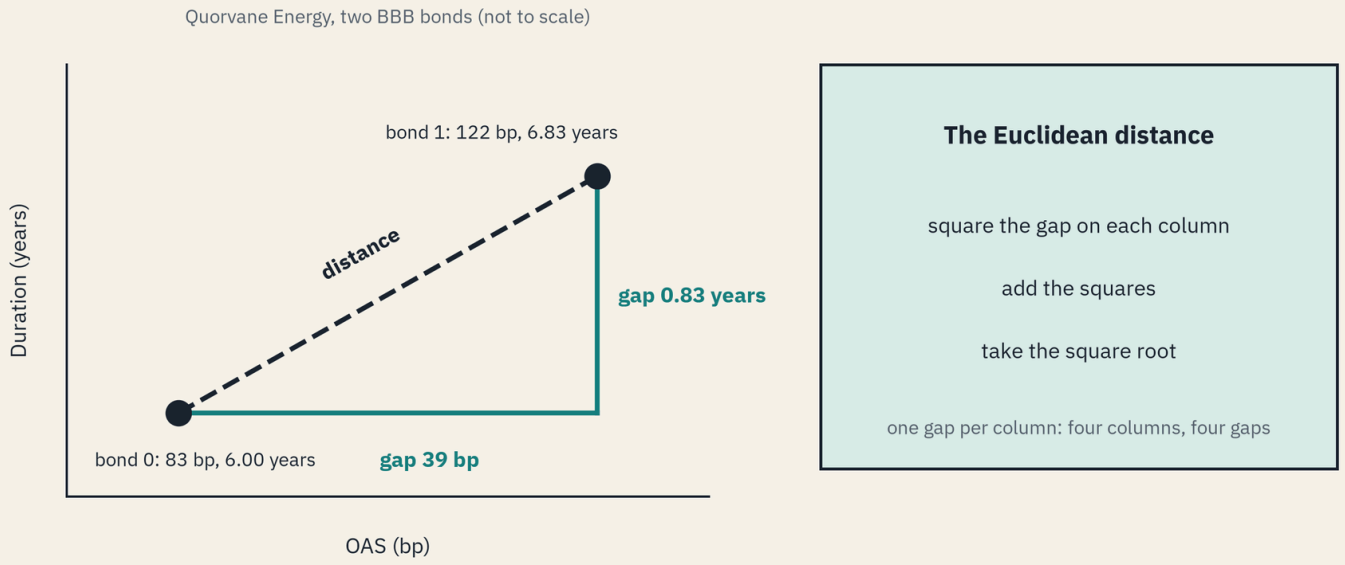


FIGURE 2 Bonds 0 and 1 sit 39 bp and 0.83 years apart, and their distance is the long side of the triangle.

WHAT A DISTANCE IS

The distance between two bonds is the straight line between them, as in Pythagoras.

Square the gap on each column, add the squares, take the square root.

This is the **Euclidean distance**, and it works for any number of columns.

SUBTRACT ONE BOND FROM THE OTHER

```
gaps = first_two.iloc[1] - first_two.iloc[0]
```

K-Means on the File's Units

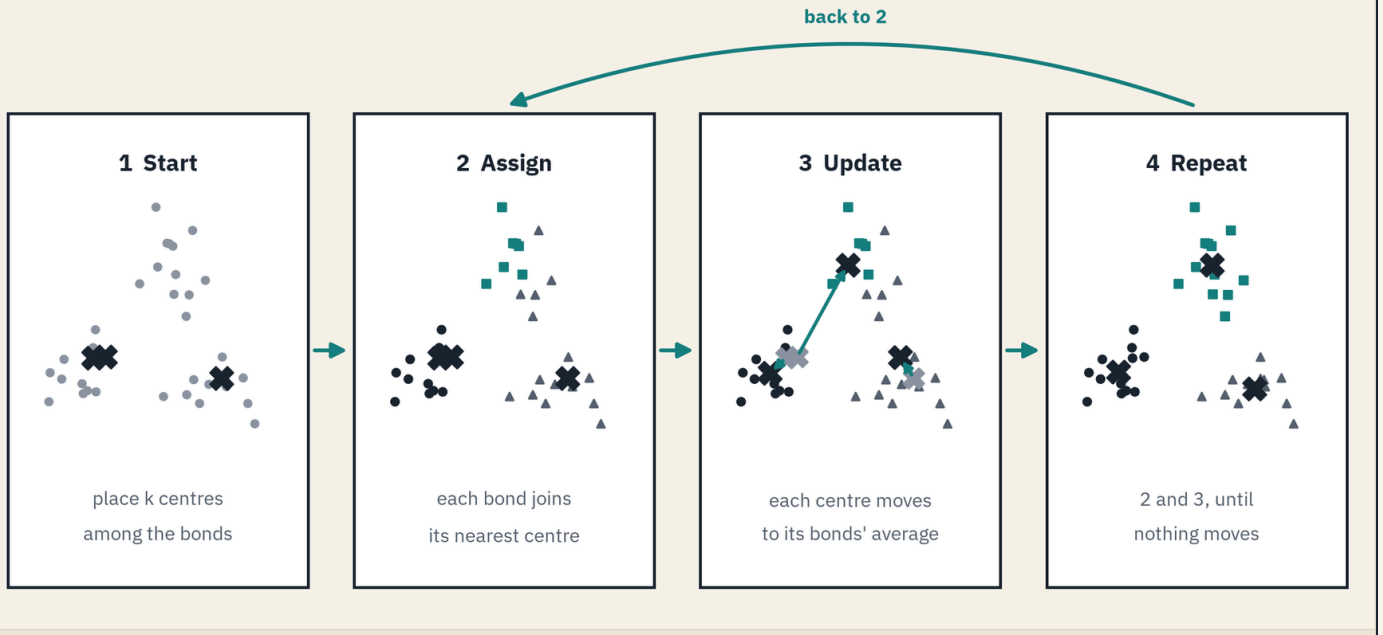


FIGURE 3 K-means places k centres, joins each bond to its nearest centre, moves each centre to its bonds' average, and repeats.

FIT FOUR CLUSTERS

```
X_raw = data[["oas", "duration", "coupon", "score"]]
kmeans_raw.fit(X_raw)
```

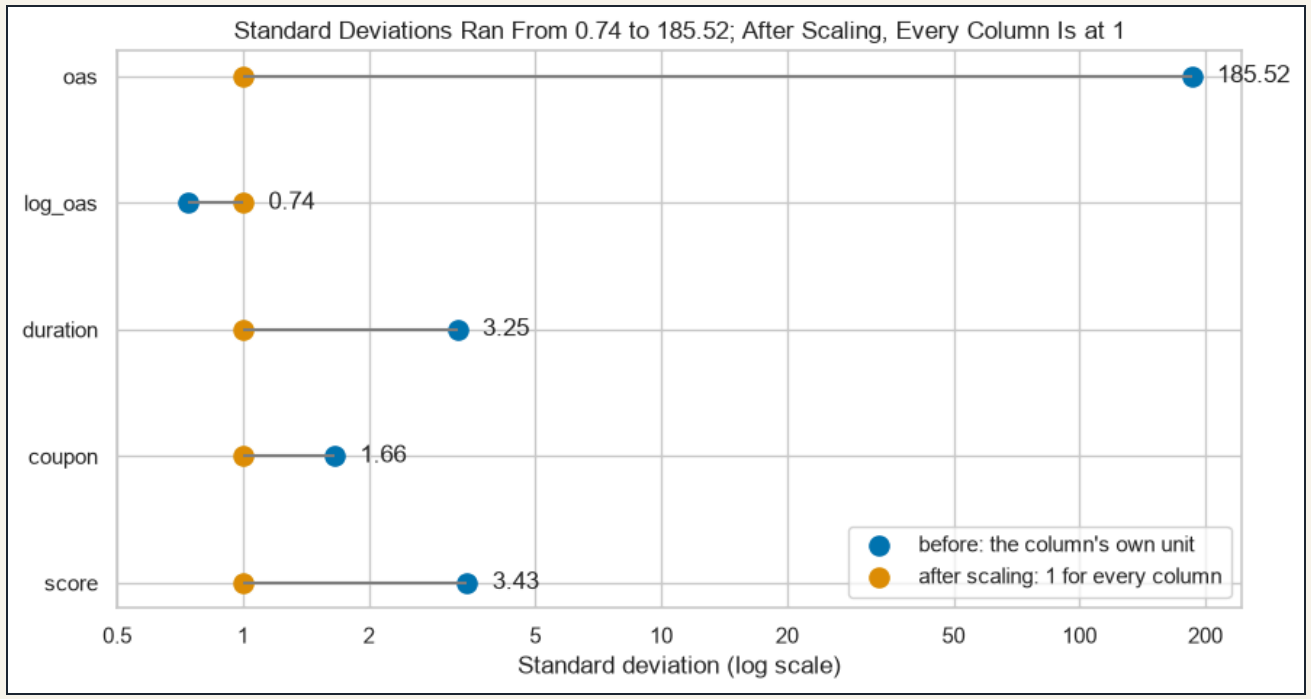
```
KMeans(n_clusters=4, n_init=10, random_state=42)
```

WHY TEN STARTS

A loop can still settle in a poor place, so **`n_init=10`** runs it from ten starts and keeps the lowest inertia.

Inertia adds up the squared distance from every bond to its own centre: the lower, the tighter the clusters.

Scaling the Columns

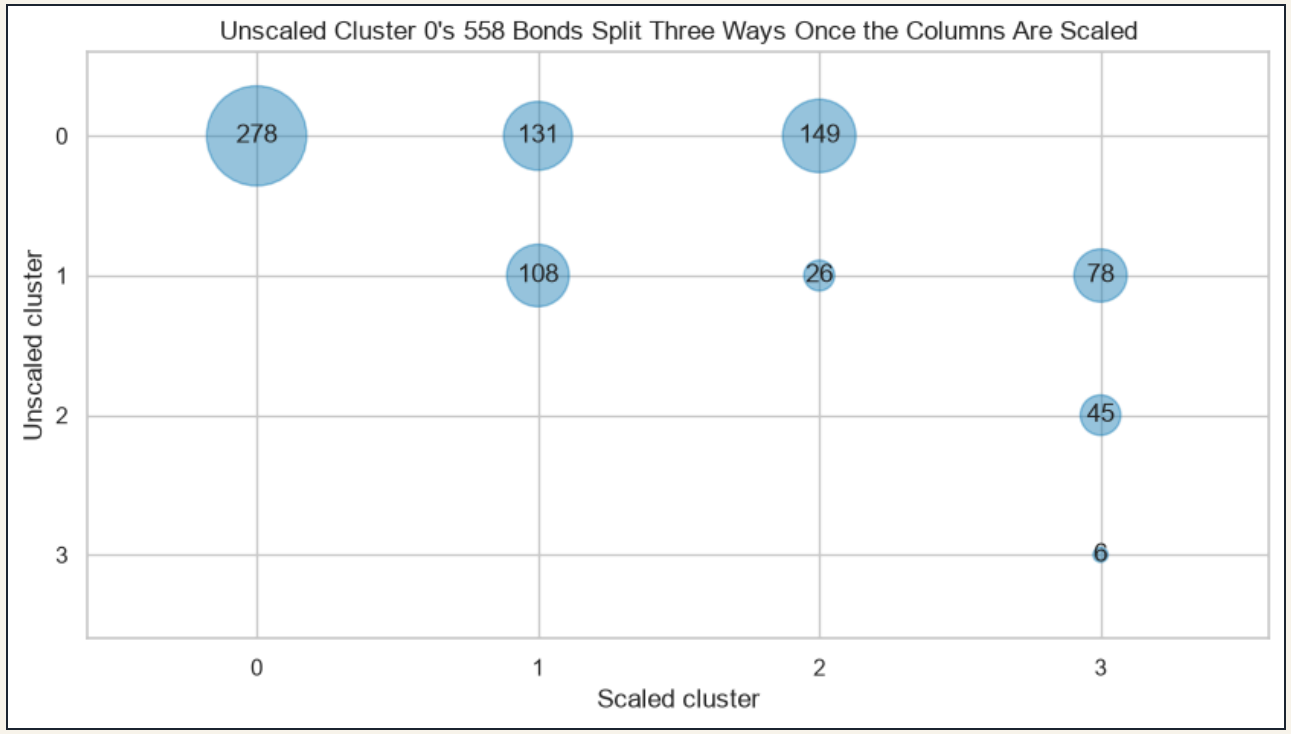


EVERY COLUMN AFTER SCALING

	LOG_OAS	DURATION	COUPON	SCORE
count	821.00	821.00	821.00	821.00
mean	-0.00	-0.00	0.00	-0.00
std	1.00	1.00	1.00	1.00
min	-2.97	-1.56	-2.63	-2.40
25%	-0.71	-0.78	-0.67	-0.66
50%	-0.08	-0.02	-0.14	-0.07
75%	0.62	0.54	0.54	0.51
max	3.74	3.09	3.11	2.55

Before scaling, OAS's standard deviation was about 57 times duration's.

Changing the Units After Scaling



OAS IN PERCENT, DURATION IN MONTHS, THEN SCALED

```
Bonds that change cluster: 0 of 821
```

WHY NO BOND MOVES

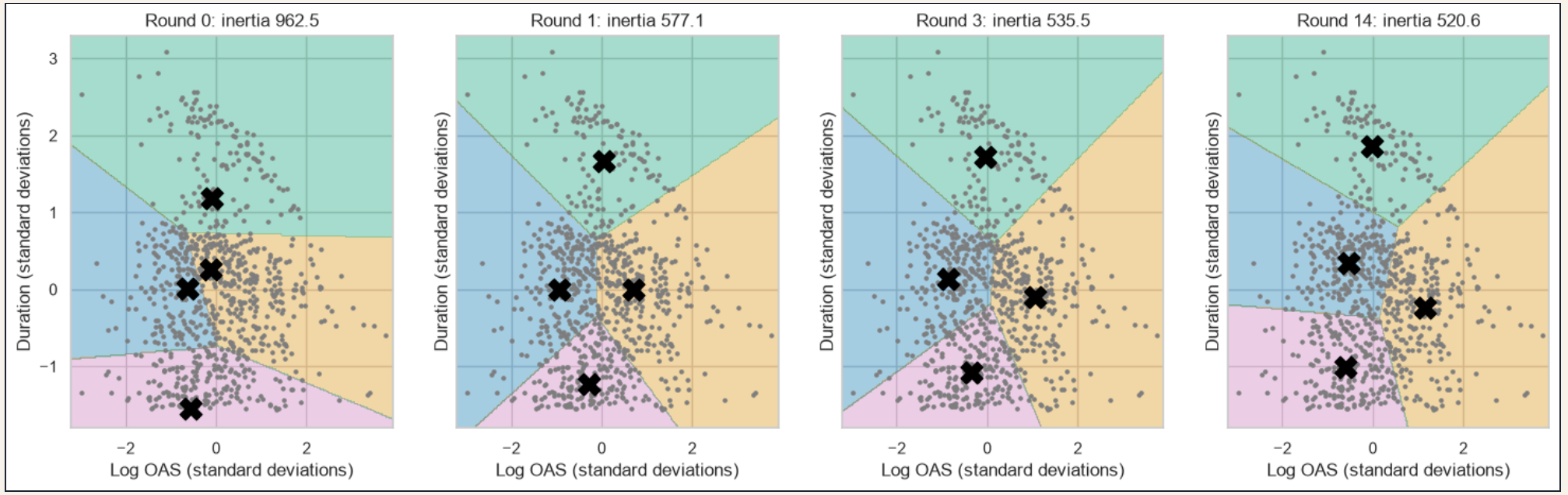
Percent shifts every log OAS by the same amount and months multiply every duration by 12, and neither changes a z-score.

READING THE CIRCLES

Unscaled cluster 0's 558 bonds split 278, 131 and 149 across three scaled clusters.

Scaled, every column has a say. Now we open the loop and watch it work.

Inside the Loop



DOES SCIKIT-LEARN AGREE WITH THE LOOP BY HAND

```
Rounds: 14  Same centres: True
```

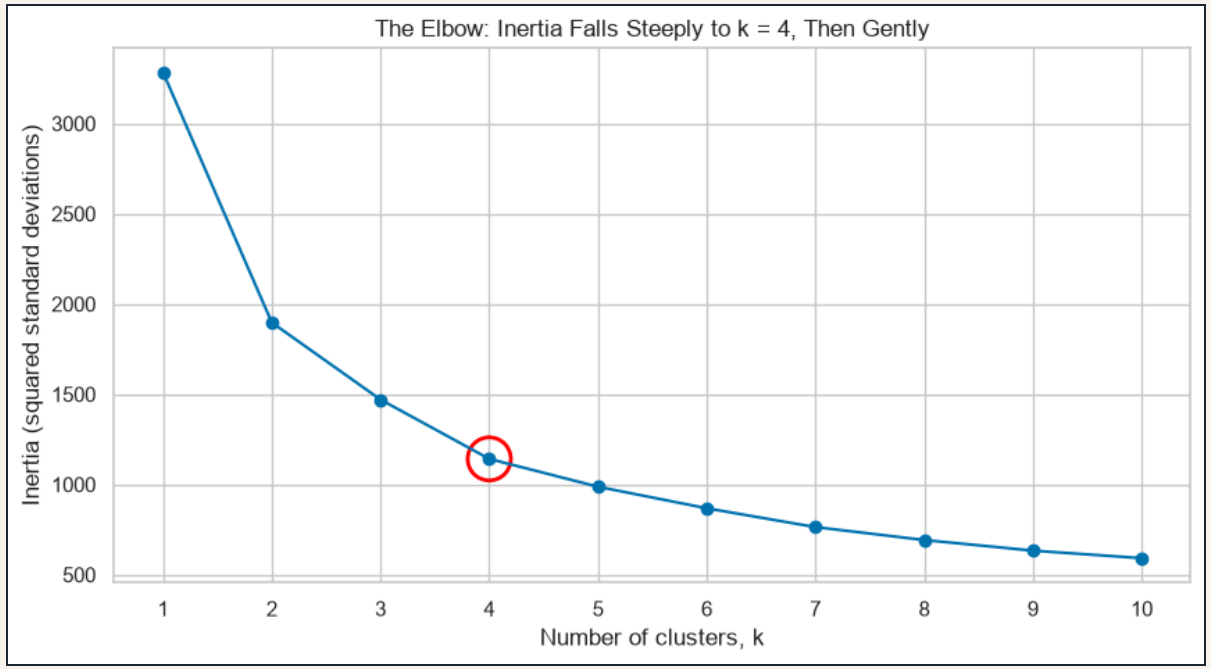
INERTIA OVER THE ROUNDS

Inertia falls from 962.5 to 577.1 in the first round, then creeps down to 520.6.

WHAT THE PICTURE SHOWS

At round 0 the four centres are the first four bonds, in a narrow strip of log OAS.

Choosing the Number of Clusters



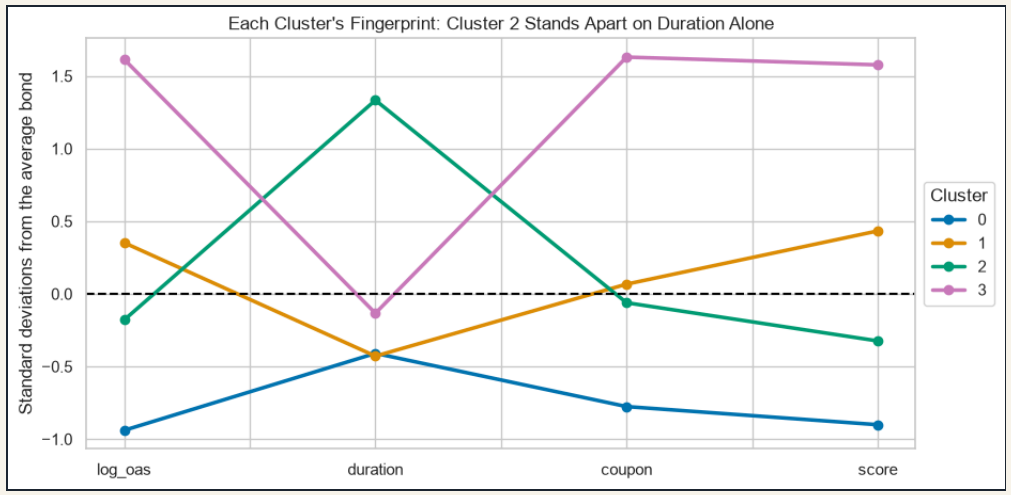
INERTIA FOR EACH K

K	INERTIA	FALL
1	3284.0	NaN
2	1897.6	1386.4
3	1470.4	427.2
4	1141.8	328.5
5	987.5	154.3
6	868.1	119.4
7	763.9	104.2
8	692.3	71.6
9	633.6	58.8
10	591.7	41.9

The curve bends gradually, so the elbow is a reading: we read it at 4, after the last big fall.

Describe each cluster in bp, years and notches.

Describing the Clusters



READING THE FINGERPRINT

Clusters 0 and 3 mirror each other: cluster 0 sits below the average bond on log OAS, coupon and score, cluster 3 well above it.

Cluster 2 sits near the average on everything but duration.

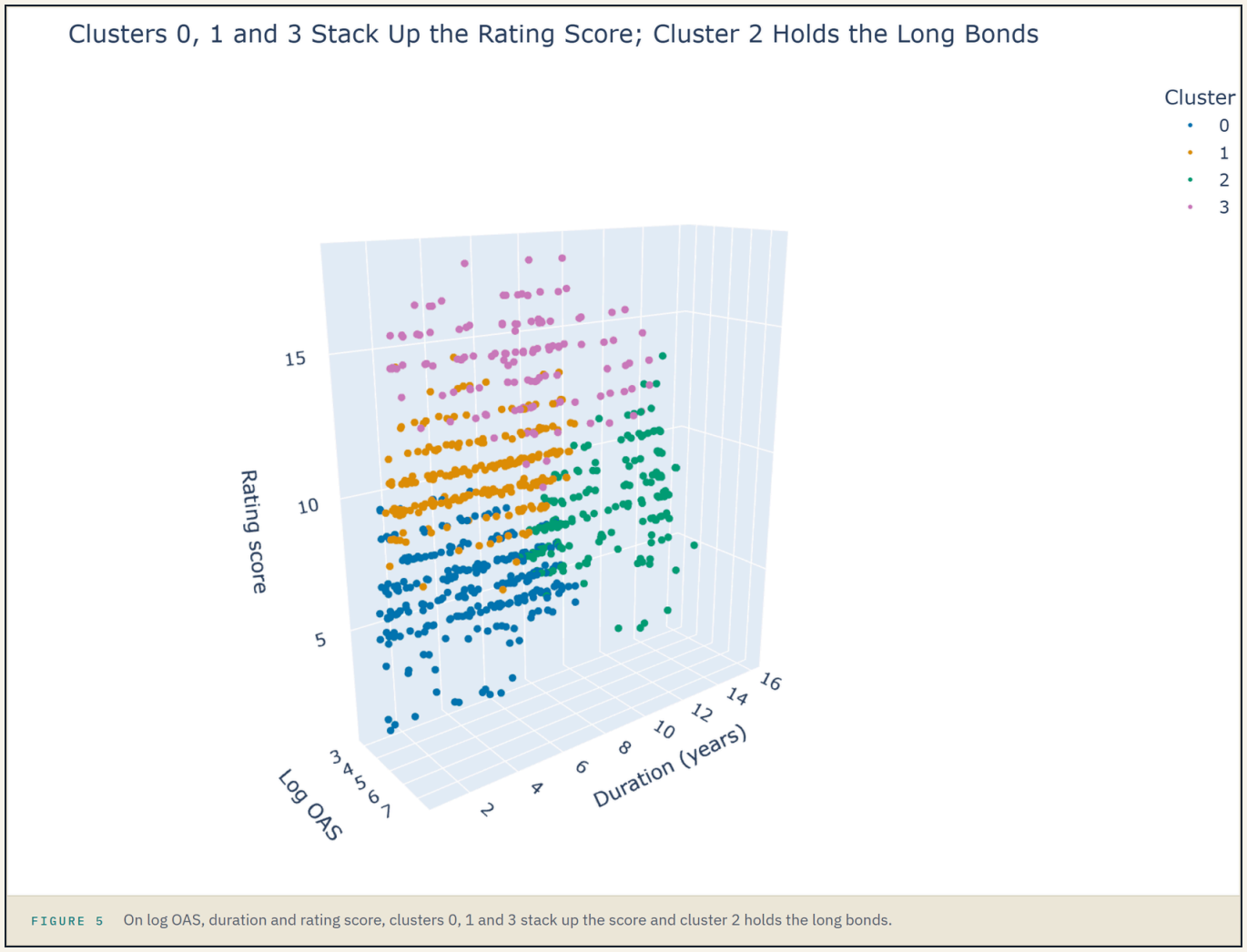
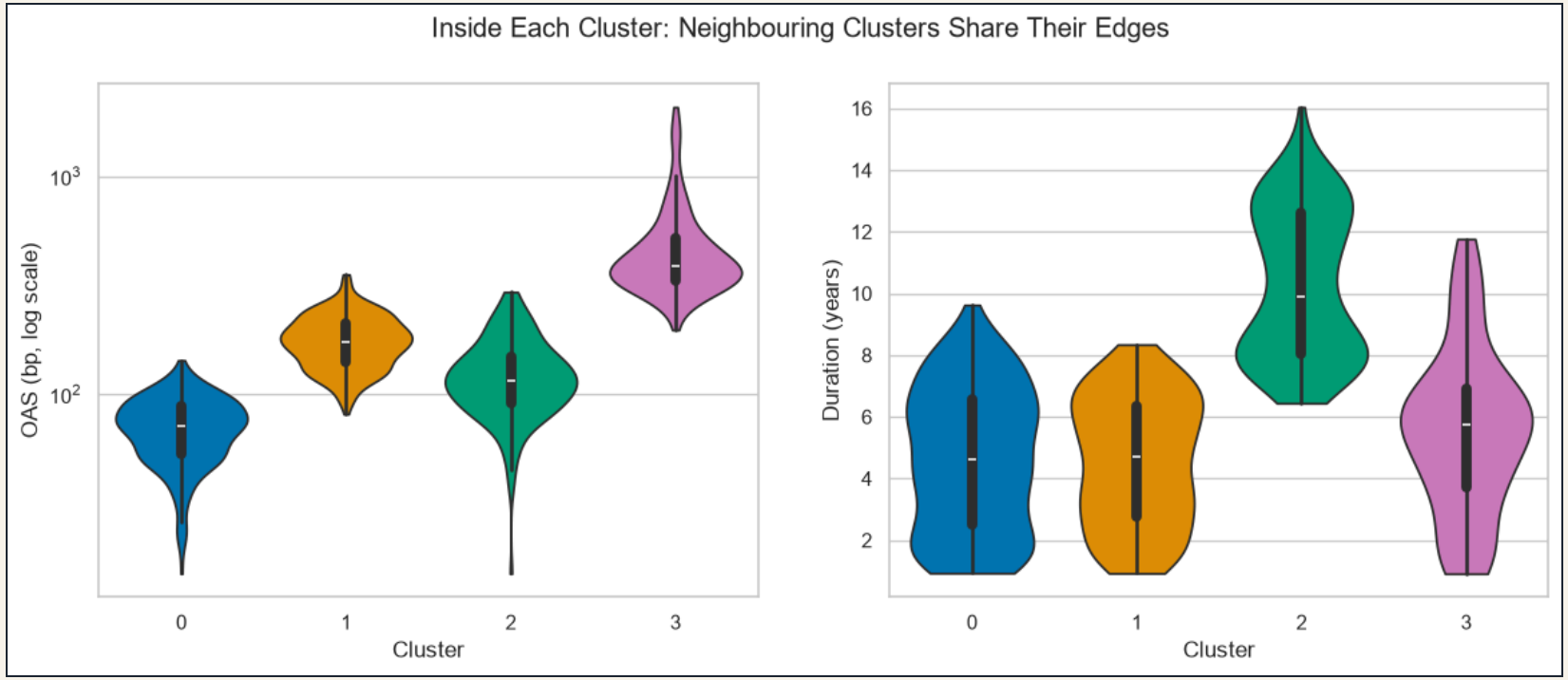


FIGURE 5 On log OAS, duration and rating score, clusters 0, 1 and 3 stack up the score and cluster 2 holds the long bonds.

Insights and Conclusion



CLUSTERS HAVE NO WALLS

The clusters overlap at their edges: some bonds in cluster 1 are wider than some in cluster 3.

A cluster is a region where similar bonds crowd, not a box with walls.

The four clusters are described in bp, years and notches: groups of similar bonds, not a ranking.

You Can Now

- Measure the distance between two bonds and see which column makes it up.
- Scale columns with **`StandardScaler`** before any method built on distance.
- Fit K-means with **`n_init`** written out, number the clusters by size, and follow its loop round by round.
- Choose k with the elbow and the silhouette, and say it is a judgement.
- Describe clusters in the file's own units without ranking them.

NEXT
WEEK 3 · DAY 3

Hierarchical Clustering and Cluster Profiling

Day 13 builds clusters from the bottom up and reads them off a tree.